

- 11 B** The net force is the centripetal force = $mr\omega^2$
For the 5.0 kg mass, the net force = $(5.0)(2.0)\omega^2 = 10\omega^2$
For the 8.0 kg mass, the net force = $(8.0)(6.0)\omega^2 = 48\omega^2$
Same ω for both masses, hence the ratio is 10: 48 = 5: 24

12 C $\frac{GM}{r^2} = \frac{G(4\pi r^3 \rho)}{r^2} = 4\pi G\rho r$